



CODECHELLA MADRID 2026

Foundations of Difference-in-Differences

Day 1: The 2×2 , Parallel Trends, and Event Studies

Scott Cunningham · Baylor University

CUNEF Auditorium · May 25–28, 2026

Welcome to Codechella Madrid 2026

Our third annual workshop here at CUNEF.

Mixtape Sessions

Scott Cunningham · Baylor

Kyle Butts · Arkansas

(at home with his new baby!)

with **Mark Anderson** (Montana State)

Dan Rees (UC3M) · **Agustín Casas** (CUNEF)

What we're here for

Practical causal inference.

The methods.

Using them well.

Our opinions, along the way.

The week

	Day 1 Monday, May 25 CUNEF Auditorium	Day 2 Tuesday, May 26 CUNEF Auditorium	Day 3 Wednesday, May 27 CUNEF Auditorium	Day 4 Thursday, May 28 CUNEF Auditorium
8:30 – 9am	Registration			
9 – 10:30am	Core DID	Covariates	Continuous	Staggered
10:30 – 11am	Coffee Break	Coffee Break	Coffee Break	Coffee Break
11am – 1pm	Covariates	Covariates	Continuous	Staggered
1 – 2:30pm	Lunch (by Research Field)	Lunch	Lunch	Lunch
2:30 – 3:30pm	<i>Dan Rees & Mark Anderson</i> Hidden Curriculum	Continuous	Staggered	Claude Code
3:30 – 5pm	<i>Dan Rees & Mark Anderson</i> Hidden Curriculum	Continuous	<i>Dan Rees & Mark Anderson</i> Hidden Curriculum	Claude Code

Today: the basics of DiD

Four moving parts.

Calculation.

Identification.

Regressions.

Falsifications.

After today

Covariates	· Mon PM + all of Tue
Continuous DiD	· Tue PM + Wed AM
Staggered DiD	· Wed PM + Thu AM
Hidden Curriculum	· Mon & Wed afternoons, with Mark & Dan
Claude Code	· Thu PM

LESSON

1. DiD's Empirical Roots

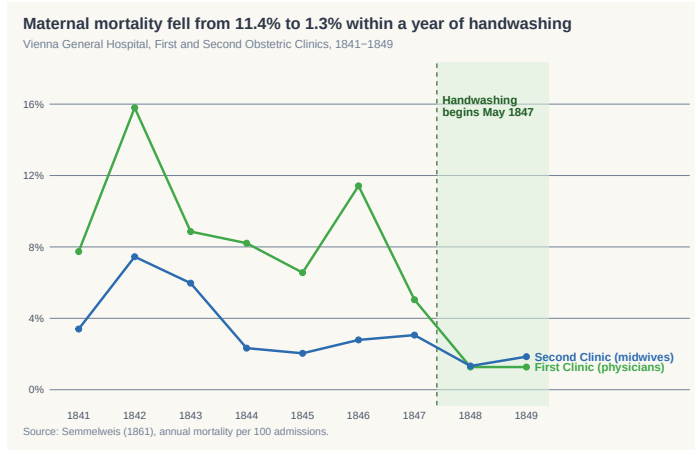
Vienna, 1840s: two clinics, two mortality rates

- Vienna General Hospital, First Obstetric Clinic
- Pregnant women were assigned to the *physicians' wing* or the *midwives' wing* by day of admission — effectively at random
- Physicians' wing: **13–18% mortality** from puerperal fever
- Midwives' wing: **about 3%**
- Women would deliver in the street rather than risk the physicians' wing

His hypothesis, and his DiD

- Semmelweis — an assistant on the wards — noticed the physicians did cadaver dissections before delivering babies. The midwives did not.
- Hypothesis: something on the physicians' hands was causing the fever.
- May 1847: he instituted **chlorinated lime handwashing for physicians only**
- Midwives served as the untreated comparison group — a DiD by hand

The data



Monthly mortality, First Obstetric Clinic. Handwashing began mid-1847.

Princeton, 1970s: Orley Ashenfelter

The Industrial Relations Section was founded 1922 — rigorous empirical labor research, the policy edge of US economics.

Orley Ashenfelter's leadership

- Long-time director. Set the tone: high-quality data, careful design, communicate to bureaucrats.
- Trained a cohort: **David Card**, **Josh Angrist**, **Bob LaLonde**, **Janet Currie**, Pischke, Dinardo, Case, others.
- Hired **Alan Krueger** as a junior colleague.

By the late 1970s, empirical labor was in trouble

- Different methods on the same question disagreed wildly
- The Princeton response: **design over modeling — estimators that mimic an RCT as closely as possible**
- Worker surveys, federal job training, massive panel regressions
- Audience was bureaucrats

Their eyes would glaze over whenever you said ‘regression.’

— Orley Ashenfelter

youtu.be/WnB3EJ8K7lg (3:53)

LESSON

2. DiD is Two Things: a Calculation and an Assumption

DiD is a research design, not a regression

A research design is an identification strategy: it reveals the invisible target parameter hidden inside a calculation.

1.

A calculation

(the 2×2)

+

2.

An assumption

(parallel trends)

The individual treatment effect

$$\delta_i = Y_i(1) - Y_i(0)$$

- One per unit. **Different** across units in general — treatment effects are heterogeneous.
- We only ever see one of $Y_i(1)$ or $Y_i(0)$. The other is the counterfactual.
- So δ_i is never directly observed at the individual level.

The ATT: a causal parameter has three ingredients

$$\text{ATT} = \underbrace{\mathbb{E}}_{\text{weighted expectations}} \left[\underbrace{Y(1) - Y(0)}_{\text{potential outcomes}} \mid \underbrace{D = 1, \text{Post}}_{\text{population}} \right]$$

- **Potential outcomes** — the contrast $Y(1) - Y(0)$
- **Population** — on whom: treated units, post-period
- **Weighted expectations** — how we average across that population ($\mathbb{E}[\cdot]$ defaults to equal weights)

The switching equation: you see one half

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

- For each unit: $Y_i(1)$ if treated, $Y_i(0)$ if not.
- The ATT asks about *both* columns. The data hands you one.

The rest of the day: strategies for recovering the ATT from the half you see.

LESSON

3. Five Regres- sions, One Number

Cheng & Hoekstra (2013): castle doctrine and homicide

The reform. “Castle doctrine” laws expanded the legal right to use lethal force in self-defense, removing the duty to retreat. Many US states passed expansions from **2005–2010**.

- **Panel:** all US states $\times$ years (state-year).
- **Outcome:** log homicide rate, FBI Supplementary Homicide Reports.
- **Treatment:** state passed an expansion of castle doctrine.
- **Comparison:** states that did not pass (never-treated).

For today: a clean 2×2 on **2005–2006**. Treated = states adopting in 2006; control = states that hadn't. *Cheng & Hoekstra (2013), “Does Strengthening Self-Defense Law Deter Crime or Escalate Violence? Evidence from Expansions to Castle Doctrine.” J. Human Resources 48(3): 821–854.*

The four cells, by hand on castle.dta

```
use castle.dta, clear
keep if year >= 2005 & year <= 2006
gen post = (year == 2006)
gen treat = (effyear == 2006)

summarize l_homicide if treat==1 & post==1 // Y11
summarize l_homicide if treat==1 & post==0 // Y10
summarize l_homicide if treat==0 & post==1 // Y01
summarize l_homicide if treat==0 & post==0 // Y00
```

$$\widehat{\text{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}) \approx \mathbf{0.10}$$

Log homicide rate ~ 10% higher in 2006-adopting states relative to non-adopters.

Regression 1: OLS with $\text{treat} \times \text{post}$

```
reg l_homicide post##treat, cluster(sid)
```

The coefficient on `1.post#1.treat` is the DiD.

$$Y = \beta_0 + \beta_1 \text{Post} + \beta_2 \text{Treat} + \beta_3 (\text{Post} \times \text{Treat}) + \varepsilon$$

The saturated specification. $\hat{\beta}_3 \approx 0.10$.

Regression 2: two-way fixed effects

```
xtset sid year  
xtreg l_homicide c.treat#c.post i.year, fe vce(cluster sid)
```

State fixed effects absorb time-invariant level differences. Year fixed effects absorb common shocks.

In the 2×2 case, the TWFE coefficient equals the DiD. Same number.

Regression 3: first-difference ΔY_i on D_i

```
keep sid year l_homicide treat  
reshape wide l_homicide, i(sid) j(year)  
gen dY = l_homicide2006 - l_homicide2005  
reg dY treat, vce(cluster sid)
```

$$\Delta Y_i = \alpha + \beta_{DD} D_i + u_i$$

- Collapse the panel: one ΔY per unit.
- Time-differencing wipes out unit fixed effects mechanically.
- Slope on treat = the DiD. Same number.

Regression 4: $(Y_{\text{treat}} - Y_{\text{ctrl}})$ on the post dummy

At each time t , take the across-group difference $D_t = \bar{Y}_{1,t} - \bar{Y}_{0,t}$. Then regress on a post indicator:

$$D_t = \alpha + \beta_{DD} \text{Post}_t + v_t$$

```
collapse (mean) l_homicide, by(treat year)
reshape wide l_homicide, i(year) j(treat)
gen Dt = l_homicide1 - l_homicide0
gen post = (year == 2006)
reg Dt post
```

- Regression 3 differences over *time*; Regression 4 differences over *groups* first.
- By Frisch–Waugh–Lovell, partialling out the two-way means runs in either order. Same number.

Regression 5: long differences — the Snow move

Compute one number per group, post minus pre, then subtract:

$$\widehat{\text{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})$$

- No regression. Two long differences. One subtraction between them.
- This is what John Snow did in 1854 with the water-company data.
- Identical to Regressions 1–4. **The arithmetic was the point all along.**

Five regressions, one number

Specification	What it does	$\hat{\beta}_{DD}$
1. OLS with <code>post##treat</code>	saturated interaction	≈ 0.10
2. Two-way fixed effects	absorb unit + time means	≈ 0.10
3. ΔY_i on D_i (first difference)	difference over time, then on D	≈ 0.10
4. D_t on <code>Post_t</code> (across-group diff)	difference over groups, then on <code>Post</code>	≈ 0.10
5. Long differences (Snow)	two means, one subtraction	≈ 0.10

*Same data, five specifications, identical to machine epsilon.
The 2×2 is the object; the regressions are just five ways to ask for it.*

LESSON

4. Decomposing the 2×2

The proof: add a zero, rearrange (five steps)

1. Start with the 2×2 :

$$\widehat{\text{DiD}} = [\mathbb{E}[Y \mid 1, \text{Post}] - \mathbb{E}[Y \mid 1, \text{Pre}]] - [\mathbb{E}[Y \mid 0, \text{Post}] - \mathbb{E}[Y \mid 0, \text{Pre}]]$$

2. Switching equation. Treated-post is $Y(1)$; the other three cells are $Y(0)$:

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}]] - [\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]]$$

3. Add a zero. Insert $+\mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}]$:

$$\begin{aligned}\widehat{\text{DiD}} &= \mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}] \\ &\quad + \mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}] - [\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]]\end{aligned}$$

4. Rearrange. First two terms = ATT; remainder = difference in $Y(0)$ trends:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid 1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta \mathbb{E}[Y(0) \mid 1] - \Delta \mathbb{E}[Y(0) \mid 0]\}}_{\text{non-parallel-trends bias}}$$

5. Read. If the bias equals zero (*parallel trends*), then $\widehat{\text{DiD}} = \text{ATT}$.

The bias term *is* selection bias — on trends, not levels

The “non-parallel-trends bias” isn’t a new concept. It’s selection bias, in a different costume.

$$\text{bias} = \underbrace{\Delta \mathbb{E}[Y(0) \mid D=1]}_{\text{how treated's } Y(0) \text{ moves}} - \underbrace{\Delta \mathbb{E}[Y(0) \mid D=0]}_{\text{how controls' } Y(0) \text{ moves}}$$

- **Cross-section selection bias:** are treated and control different in *levels* of $Y(0)$?
- **DiD’s selection bias:** are treated and control different in *trends* of $Y(0)$?
- Different levels are fine. Unit fixed effects soak them up.
- What can’t differ is the *rate* at which $Y(0)$ moves across groups.

The trade DiD makes vs. randomization

RCTs: treated and controls identical in everything — levels and trends.

DiD: identical only in *trends* of $Y(0)$. **Cheaper assumption. Weaker identification.**

Case 1: NA holds, never-treated comparison

Switching equation. Treated-post is $Y(1)$; the other three cells are $Y(0)$.

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}]] - [\mathbb{E}[Y(0)|0, \text{Post}] - \mathbb{E}[Y(0)|0, \text{Pre}]]$$

Add the zero. Insert $+\mathbb{E}[Y(0)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Post}]$. Rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta\mathbb{E}[Y(0)|1] - \Delta\mathbb{E}[Y(0)|0]\}}_{\text{non-PT bias} = 0 \text{ under PT}}$$

Numerical example

ATT = 10, PT bias = 0. $\widehat{\text{DiD}} = 10$. ✓

Case 2: NA fails — anticipation contaminates Y_{Pre}

Suppose the treated group anticipates and adjusts *before* the policy lands. Their pre-period outcome contains some $Y(1)$ response: $\mathbb{E}[Y|1, \text{Pre}] = \mathbb{E}[Y(0)|1, \text{Pre}] + \text{ATT}_{\text{pre}}$.

Same +0 trick, propagated:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}_{\text{Post}}}_{\text{what we want}} + \underbrace{\text{non-PT bias}}_{=0} - \underbrace{\text{ATT}_{\text{pre}}}_{\text{anticipation}}$$

Numerical example

$\text{ATT} = 10, \text{PT bias} = 0, \text{ATT}_{\text{pre}} = 3. \quad \widehat{\text{DiD}} = 10 - 3 = 7. \text{ biased down}$

If treatment effects are roughly constant, DiD can collapse to zero.

Case 3: the control group also receives treatment in Post

Now the comparison group's Post outcome is also $Y(1)$, not $Y(0)$. Switching equation:

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}]] - [\mathbb{E}[Y(1)|0, \text{Post}] - \mathbb{E}[Y(0)|0, \text{Pre}]]$$

Add two zeros — one missing counterfactual on each side — and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta\mathbb{E}[Y(0)|1] - \Delta\mathbb{E}[Y(0)|0]\}}_{\text{non-PT bias} = 0 \text{ under PT}} - \underbrace{\mathbb{E}[Y(1) - Y(0)|0, \text{Post}]}_{\text{ATT}_{D=0, \text{Post}} \text{ (spillover)}}$$

Numerical example

$\text{ATT} = 10$, $\text{PT bias} = 0$, $\text{ATT}_{D=0, \text{Post}} = 4$. $\widehat{\text{DiD}} = 10 - 4 = 6$. **biased toward zero**

Case 4: the comparison group is *always* treated

The comparison group was treated *before* our data starts. Both their Pre and Post outcomes are $Y(1)$, not $Y(0)$. Switching equation:

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}]] - [\mathbb{E}[Y(1)|0, \text{Post}] - \mathbb{E}[Y(1)|0, \text{Pre}]]$$

Add two zeros — $\pm \mathbb{E}[Y(0)|0, \text{Post}]$ and $\pm \mathbb{E}[Y(0)|0, \text{Pre}]$ on the comparison side — and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta \mathbb{E}[Y(0)|1] - \Delta \mathbb{E}[Y(0)|0]\}}_{\text{non-PT bias} = 0 \text{ under PT}} - \underbrace{\Delta \text{ATT}_{D=0}}_{\text{change in comparison TE}}$$

Numerical example

$\text{ATT} = 10$, $\text{PT bias} = 0$, $\Delta \text{ATT}_{D=0} = 2$. $\widehat{\text{DiD}} = 10 - 2 = 8$. **biased toward zero**

Four cases, four failures of the punchline. The +0 trick still works — only the zero you add changes.

Summary: four ways the 2×2 decomposes

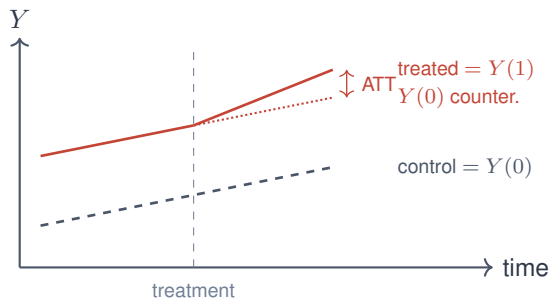
What holds	$\widehat{\text{DiD}} =$
NA, never-treated control	$\text{ATT} + \text{PT-bias}$
PT, no NA, never-treated control	$\text{ATT}_{\text{Post}} - \text{ATT}_{\text{Pre}}$
PT, NA, always-treated control	$\text{ATT} - \Delta \text{ATT}_{D=0}$
PT, NA, SUTVA fails (spillover)	$\text{ATT} - \text{ATT}_{D=1, \text{Post}}$

All four derivable the same way: write down the 2×2 , apply the switching equation, add the right series of zeros.

LESSON

5. Card and Krueger: Parallel Trends, Visualized

Parallel trends, drawn



$$\mathbb{E}[Y(0) \mid D=1, \text{Post}] - \mathbb{E}[Y(0) \mid D=1, \text{Pre}] = \mathbb{E}[Y(0) \mid D=0, \text{Post}] - \mathbb{E}[Y(0) \mid D=0, \text{Pre}]$$

*A 2×2 of $Y(0)$ terms. **If it isn't four $Y(0)$ s, it isn't the PTA.***

Under PT, the 2×2 hits the ATT

With (i) **No Anticipation**, (ii) **Parallel Trends in $Y(0)$** , and (iii) a clean (never-treated) comparison group:

$$ATT_{\text{post}} = \underbrace{\mathbb{E}[\Delta Y \mid D = 1]}_{\text{treated change}} - \underbrace{\mathbb{E}[\Delta Y \mid D = 0]}_{\text{control change}}$$

- The 2×2 calculation equals the population's ATT.
- Four averages. Three subtractions. One target parameter.

This is the punchline. Everything that follows is variations on this theme.

New Jersey raises its minimum wage. Pennsylvania doesn't.

- **April 1992**: New Jersey raises its minimum wage from \$4.25 to \$5.05 — a 19% increase.
- Pennsylvania, just over the river, stays at \$4.25.
- Card and Krueger survey 410 fast-food restaurants on both sides of the border, before (Feb 1992) and after (Nov 1992).
- The natural 2×2 : **treated = NJ, control = PA, pre/post the policy.**

Card & Krueger, American Economic Review, 1994.

What Card and Krueger *found*

Employment per restaurant	Pre (Feb 1992)	Post (Nov 1992)	Change
NJ (treated)	20.44	21.03	+0.59
PA (control)	23.33	21.17	−2.16
DiD (NJ – PA)			+2.76 jobs/restaurant

- Employment in NJ fast food *rose* slightly, relative to PA, after the wage hike.
- Standard error puts the effect roughly indistinguishable from zero — but the sign is the opposite of what theory predicted.
- **No measurable disemployment effect.** The textbook had failed a real test.

The reaction was personal

No self-respecting economist would claim that increases in the minimum wage increase employment. Such a claim becomes equivalent to a denial that there is even minimal scientific content in economics.

— James Buchanan (Nobel 1986), *WSJ*, April 1996

It cost me a lot of friends. People that I had known for many years became very angry or disappointed. They thought that in publishing our work we were being traitors to the cause of economics as a whole.

— David Card, interview

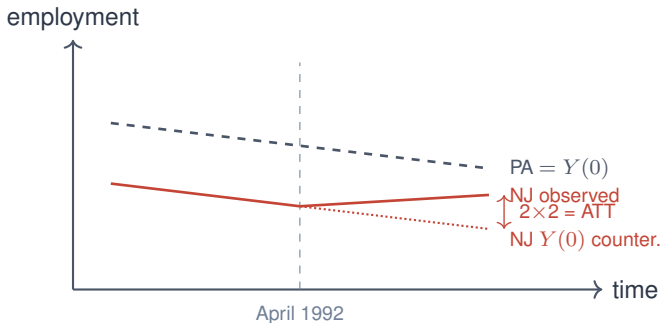
The price of a credible empirical paper that contradicted the consensus. Krueger paid more.

Hear it from them — three short interview clips

- **Orley Ashenfelter on the profession's reaction** (start 31:20)
youtu.be/M0tbuRX4eyQ?t=1882
- **Card on the hostility from peers** (1:07–1:10)
youtu.be/1soLdywFb_Q?t=4020
- **Card on what happened to Alan Krueger** (1:16–1:17)
youtu.be/1soLdywFb_Q?t=4556

The DiD literature did not arrive politely. It was earned through paper-by-paper combat, much of it personal.

Parallel trends visualized: the control *is* the counterfactual



The dotted line is what NJ would have done absent the policy. We never see it.

PT lets PA's slope stand in for it.

Aside: there's also an *across-group* PT we never name

The usual statement (within-group, across-time)

The comparison group's $Y(0)$ trend stands in for what the treated group's $Y(0)$ trend would have been.

The same assumption, stated from the other side (across-group, within-time)

The treated group's $Y(0)$ trend (had they not been treated) would equal the comparison group's $Y(0)$ trend.

- Two phrasings of one assumption: $Y(0)$ trends are common across groups.
- Event-study β_k vs $t = -1$ computes the within-group across-time piece.
- Regression 3 (ΔY on D) and Regression 4 (D_t on Post) are FWL mirrors — same PT from the two directions.

ASIDE — drop in or drop out if behind schedule.

Pre-trends are the gut check — not a proof of PT

- Card–Krueger held up because in the pre-period, NJ and PA fast-food employment moved together.
- If NJ employment had already been diverging before April 1992, the +2.76 estimate would have been mostly trend, not policy.
- Flat pre-trends → evidence (not proof) the design holds.
- **The 2×2 was credible because of the pre-period parallelism — not because of the regression coefficient.**

More later (Roth, Roth–Sant’Anna): when pre-trends look flat but the test still over-rejects.

LESSON

6. Open the Sheet

Open the Sheet: stability and trends, by hand

Live demo

Compute the four cells, three subtractions, and the by-hand event study — in a spreadsheet.

docs.google.com/spreadsheets/d/1onabpc14JdrGo6NFv0zCWo-nuWDLLV2L1qNogDT9SBw

Four tabs in sequence: Potential Outcomes $\rightarrow$ 2X2 $\rightarrow$ Balancing $\rightarrow$ 2XT #1.

LESSON

7. Standard Errors

Three flavors of standard error

- **Vanilla (homoskedastic)**: assumes $\text{Var}(\varepsilon_i \mid X_i) = \sigma^2$ for all i .
- **Robust (Eicker–Huber–White, HC)**: each residual speaks for itself — handles heteroskedasticity.
- **Cluster-robust (CRVE)**: groups residuals by cluster; within-cluster errors correlate freely.

Same point estimate, three different SEs. The SE you pick is a claim about the error structure.

HC1 numerically: each $\hat{e}_i^2$ weighted by leverage

Six observations. Three pairs at $\tilde{x} \in \{-2, 0, +2\}$. Fitted $\hat{\beta}_1 = 1$; residuals shown.

i	cluster g	$\tilde{x}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$\tilde{x}_i^2 \cdot \hat{e}_i^2$
1	1	-2	+0.5	0.25	1.00
2	1	-2	+0.5	0.25	1.00
3	2	0	-1.0	1.00	0.00
4	2	0	-1.0	1.00	0.00
5	3	+2	+0.5	0.25	1.00
6	3	+2	+0.5	0.25	1.00
meat =					4.00

With $n = 6$, $k = 2$, and $S_{xx} = \sum_i \tilde{x}_i^2 = (-2)^2 + (-2)^2 + 0 + 0 + 2^2 + 2^2 = 16$:

$$\widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}_1) = \underbrace{\frac{n}{n-k}}_{\text{bread}} \cdot \underbrace{\frac{1}{S_{xx}}}_{\text{bread}} \cdot \underbrace{\text{meat}}_{4.00} = \frac{6}{4} \cdot \frac{4.00}{16 \cdot 16} = \frac{6}{4} \cdot \frac{4.00}{256} = 0.02344$$

$$\Rightarrow \text{SE}_{\text{HC1}} = \sqrt{0.02344} \approx 0.153$$

Obs 3–4 contribute zero because $\tilde{x} = 0$ kills the leverage weight. **HC1 treats all 6 obs as independent.**

Why HC-robust barely helps in DiD

The HC correction reweights residuals by $X_i X_i'$. In a DiD with binary $D_i \in \{0, 1\}$ and roughly balanced groups:

- Treated and control units have similar *leverage*: each side carries $\approx$ half the weight in $X'X$.
- Residual variance is roughly symmetric across the two groups (binary D doesn't generate a fan shape in $\hat{e}^2$).
- The weighted sum $\sum_i \hat{e}_i^2 X_i X_i'$ ends up not very different from $\hat{\sigma}^2(X'X)$.

Consequence

$SE_{HC} \approx SE_{vanilla}$ for typical DiD specifications.

The over-rejection problem is not heteroskedasticity. It's something else.

CRVE numerically: same 6 obs, group scores, then square

Same 6 observations. Now compute $s_g = \sum_{i \in g} \tilde{x}_i \hat{e}_i$ for each cluster, then s_g^2 :

i	g	$\tilde{x}_i$	$\hat{e}_i$	$\tilde{x}_i \hat{e}_i$			
					g	s_g	s_g^2
1	1	-2	+0.5	-1			
2	1	-2	+0.5	-1	1	-2	4
3	2	0	-1.0	0	2	0	0
4	2	0	-1.0	0	3	+2	4
5	3	+2	+0.5	+1	meat =		8
6	3	+2	+0.5	+1			

With $G = 3$, $n = 6$, $k = 2$, and $S_{xx} = \sum_i \tilde{x}_i^2 = 16$, so $S_{xx}^2 = 256$:

$$\widehat{\text{Var}}_{\text{CRVE}}(\hat{\beta}_1) = \frac{G}{G-1} \cdot \frac{n-1}{n-k} \cdot \frac{\text{meat}}{S_{xx}^2} = \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{8}{256} = 0.05859 \Rightarrow \text{SE}_{\text{CRVE}} \approx 0.242$$

HC1: 0.153 vs. **CRVE: 0.242** (**58% larger**). Same data. The cluster cross-terms changed the *meat* from 4 to 8.

Bertrand, Duflo & Mullainathan (QJE 2004): the placebo-law test

“How Much Should We Trust Differences-in-Differences Estimates?”

- BDM generate *placebo laws*: assign random treatment dates to states in a real CPS panel.
- True effect by construction: $\beta = 0$. So $\Pr(\text{reject } H_0)$ should equal $\alpha = 5\%$.
- **Vanilla SE**: rejection rate $\approx 45\%$. Wildly over-rejects.
- **Heteroskedasticity-robust (HC1)**: rejection rate *also* $\approx 45\%$. **HC barely moves the needle.**
- **Cluster-robust at state level**: rejection rate $\approx 5\%$. The fix.

Why HC doesn't save you

Binary, balanced D + within-state serial correlation. The het-correction has nothing to do with the actual problem. **Clustering is doing all the work.**

LESSON

8. Population Weighting

Ten people, two counties

Person	$Y(0)$	$Y(1)$	δ_i	County
Alan	0	1	1	1
Betty	1	1	0	1
Chad	1	1	0	1
Daniel	0	0	0	1
Edith	0	1	1	1
Frank	0	1	1	1
George	0	0	0	1
Hank	0	1	1	1
Ida	1	0	-1	2
Janet	1	0	-1	2

County 1: eight people. County 2: two people. Compute the average δ for each county, and across all ten. Same answer?

Same data, two weights, two parameters

- **Within counties:** County 1 avg $\delta = \frac{1}{8}(1+0+0+0+1+1+0+1) = +0.5$.
County 2 avg $\delta = \frac{1}{2}(-1+(-1)) = -1.0$.
- **ATE for the average person:** $\frac{1}{10}(8 \cdot 0.5 + 2 \cdot (-1)) = +0.2$.
- **ATE for the average county:** $\frac{1}{2}(0.5 + (-1)) = -0.25$.

*Same potential outcomes. **Different signs.***

Whichever you report depends on what you're trying to answer.

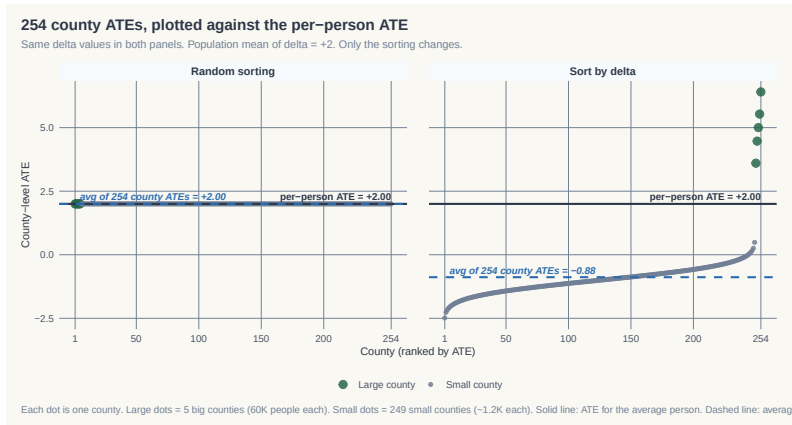
Two parameters, two answers

Scale the example up: **Texas, 254 counties, 30M people**. 5 biggest counties hold roughly half the population. Heterogeneous δ_i across people, with the high- δ folks concentrated in the 5 big counties and the low- δ folks in the 249 small ones.

- **Per-person ATE:** $\frac{1}{N} \sum_i \delta_i$ — weight every person by $1/N$. **Answer: +2.**
- **Avg of county ATEs:** $\frac{1}{254} \sum_c \bar{\delta}_c$ — weight every county by $1/254$. **Answer: -0.88.**

Same data. Two weights. Two parameters. Sometimes two signs.

Sorting on δ can flip the sign of your answer



Random: every county's ATE equals the population's. δ -sort: per-person line at +2, avg of county ATEs flips to -0.88 .

Random sorting is the benchmark — rarely reality. People choose where to live, and on traits that move δ .

The weighted mean is a covariance correction

For any outcome Y and any weight S :

$$\tilde{\mathbb{E}}[Y] \equiv \frac{\mathbb{E}[SY]}{\mathbb{E}[S]} = \mathbb{E}[Y] + \frac{\text{Cov}(S, Y)}{\mathbb{E}[S]}$$

- Weighted mean = unweighted mean + a **covariance correction**.
- $\text{Cov}(S, Y) > 0 \Rightarrow$ weighted exceeds unweighted; $< 0 \Rightarrow$ smaller.
- Ten-people table: weight by county size $S \in \{8, 2\}$ and compute $\tilde{\mathbb{E}}[\delta]$.

Stata

```
gen size = 8 if county==1
replace size = 2 if county==2
summarize delta [aweight=size]
* weighted mean = +0.2
summarize delta
* unweighted mean = -0.25
```

R

```
df$size <- ifelse(df$county==1, 8, 2)
weighted.mean(df$delta, df$size)
# weighted mean = +0.2
mean(df$delta)
# unweighted mean = -0.25
```

Pick a target parameter

- Three weights. Three signs. Three answers.
- The choice is about *which causal effect you actually want*.
- **Person-weighted** for aggregate human welfare: how many people are affected, by how much?
- **Unit-weighted** when locations have autonomy: each policy unit chooses independently — the unit *is* the agent.

Today's target: the ATT.

The average effect for the units that actually received the treatment.

LESSON

9. Event Studies: PT Made Falsifiable

Event studies are PT falsifications. What is a falsification?

- Popperian reasoning: you don't confirm a theory — you try to falsify it.
- A rival theory earns scrutiny by making a *prediction you can check*.
- Claim a confounder. Derive what *else* it would do. Test that.
- If the prediction holds → you haven't confirmed the rival, but you haven't ruled it out either.
- If the prediction fails → the rival weakens.

Popperian asymmetry. Failing to reject the rival does not confirm you. But rejecting the rival's prediction is real evidence.

Two forms of falsification

- **Same outcome, different group.** A group your treatment shouldn't reach, but the rival would.
Min wage on low-wage employment? Re-run on higher-wage workers.
- **Same group, different outcome.** An outcome your treatment shouldn't move, but the rival would.
Cigarette tax on smoking-related hospitalizations? Re-run on appendicitis and fractures.

PT falsifications come in both forms. Event studies are one (same group, different “outcome” — a period when no treatment happened).

Famous falsification 1: obesity contagion

- Christakis & Fowler (2007): **obesity is contagious** in social networks.
- Cohen-Cole & Fletcher (2008): re-run the same model on **height, acne, headaches** — same Framingham network, same identification.
- All three also show “contagion.”

We have not been able to become taller by having tall friends.

— Cohen-Cole & Fletcher (2008)

Same group, different outcome. The result isn't necessarily wrong — the empirical model is.

Famous falsification 2: milk addiction

- Becker–Murphy (1988): serial correlation in cigarette / alcohol $\Rightarrow$ “**rational addiction.**”
- Auld & Grootendorst (2004, *J. Health Econ.*): apply the same estimator to **milk, eggs, oranges.**
- All three look “rationally addictive.”

*Same group, different outcome. Nobody is rationally addicted to milk. **The identification — not the goods — is what's broken.***

Point: PT makes predictions beyond the event study. Placebo groups and placebo outcomes are the other two forms.

Parallel trends is about $Y(0)$, not “trends” abstractly

The parallel-trends assumption is a statement about an unobserved object:

$$\mathbb{E}[Y(0)_t - Y(0)_{t-1} \mid D=1] = \mathbb{E}[Y(0)_t - Y(0)_{t-1} \mid D=0]$$

- PT is the counterfactual untreated outcome's trend. **Not** a statement about OBSERVED Y trends in general.
- Observed Y for treated post-treatment is $Y(1)$, not $Y(0)$. The PT assumption can't be read off it.
- The confusion comes from saying “trends are parallel.” What we mean is “**untreated potential outcome trends are parallel.**”

Why this matters

We never see treated $Y(0)$ post-treatment. Pre-treatment data is the only window onto whether a similar assumption *could* hold post. That's where event studies come in.

Event studies: pre-treatment placebos for PT

The falsification logic. Suppose some confounder both shifts $Y(0)$ trends *and* is associated with treatment timing. That's a PT violation.

- If a similar confounder structure existed *before* treatment, we should see treated and control Y trends diverge in the pre-period too.
- Estimate placebo “effects” at $t = -2, -3, \dots, -q$ where there is no treatment. If those coefficients are large or trending, we've rejected the assumption that drives the post-period.

Popperian asymmetry

Flat pre-trends *fail to reject* PT — they do not prove it.

Sloped pre-trends *reject* the design.

Pre-treatment placebos are a falsification, not a proof.

An event study is the 2×2 viewed in motion

- Instead of one pre-period and one post-period, use *many*.
- Estimate a separate effect at each event-time relative to treatment.
- Plot them: $\hat{\beta}_{-q}, \dots, \hat{\beta}_{-2}, \hat{\beta}_{-1}, \hat{\beta}_0, \hat{\beta}_{+1}, \dots, \hat{\beta}_{+q}$.
- The graph tells two stories: **pre-trends** (left of zero) and **dynamics** (right of zero).

Same arithmetic as the 2×2 , repeated against a common reference period.

Every coefficient is a 2×2 against $t = -1$

For each event-time k (with $t = -1$ as the reference):

$$\hat{\beta}_k = (\bar{Y}_k^T - \bar{Y}_{-1}^T) - (\bar{Y}_k^C - \bar{Y}_{-1}^C)$$

- Two long differences, one subtraction — the 2×2 again, with $t = -1$ as the pre-period.
- Pre-period $\hat{\beta}_k$ ($k < -1$): asks whether trends were already diverging.
- Post-period $\hat{\beta}_k$ ($k \geq 0$): the dynamic ATTs.

Drop $k = -1$ as the reference. Multicollinearity won't let you keep all the dummies.

The saturated OLS specification

$$Y_{it} = \alpha_i + \tau_t + \sum_{t \neq t_0 - 1} \beta_t \cdot (D_i \cdot \mathbf{1}\{\text{year} = t\}) + \varepsilon_{it}$$

- α_i — unit fixed effects (absorb levels).
- τ_t — calendar-year fixed effects (absorb common shocks).
- One interaction per calendar year, omitting $t_0 - 1$ (the year just before treatment) as the reference.
- Each β_t is the 2×2 of the previous slide: (treated change from $t_0 - 1$ to t) — (control change from $t_0 - 1$ to t).

In a $2 \times T$ design the calendar year is the event-time. We save k for staggered treatment, where cohorts have different treatment dates.

Computing $\hat{\beta}_{2004}$ on castle.dta — by hand vs. OLS

Castle doctrine cohort treated in 2006. Use 2005 as the omitted reference. The 2004 coefficient is one event-study 2×2 .

By hand

```
su l_homicide if treat==1 & year==2004
* mean = 1.802504
su l_homicide if treat==1 & year==2005
* mean = 1.775398
su l_homicide if treat==0 & year==2004
* mean = 1.254353
su l_homicide if treat==0 & year==2005
* mean = 1.282884

di (1.802504-1.775398) - (1.254353-1.282884)
* 0.05564
```

$$\hat{\beta}_{2004} = \mathbf{0.05564}$$

Via OLS in Stata

```
reg l_homicide ///
    treat##ib2005.year, ///
    cluster(state)

di _b[1.treat#2004.year]
* 0.05564
```

$$\hat{\beta}_{2004} = \mathbf{0.05564}$$

Cluster-robust SE: 0.060. Match to 5 decimals.

Every event-study coefficient is a 2×2 relative to $t = -1$ (here, 2005).

LESSON

10. Five Pieces of a Strong DiD

Medicaid expansion and near-elderly mortality

Miller, Johnson & Wherry (2021), *Quarterly Journal of Economics*.

- Did the ACA Medicaid expansion (2014) reduce mortality among the **near-elderly** (ages 55–64)?
- Comparison: states that expanded vs. states that didn't, before vs. after 2014.
- Headline finding: $\approx 9.2\%$ **reduction** in near-elderly mortality.

Why *near-elderly*?

Everyone 65+ has Medicare — Medicaid expansion is irrelevant for them. The elderly become a natural **placebo group** living in the same economies.

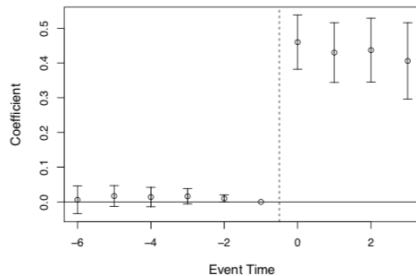
Today: walk through how they make the case using the five pieces of a strong DiD.

Five pieces of a strong DiD — start with bite

1. **Bite** — the policy actually changed what it was supposed to change.
2. **Event study** — pre/post coefficients around treatment.
3. **Falsification** — run a rival theory on a placebo group or outcome.
4. **Main result** — the DiD on the actual outcome.
5. **Mechanism** — the causal pathway from D to Y .

If the policy didn't actually move what it was supposed to move, nothing downstream is attributable to it.

Bite #1: Medicaid take-up jumped in expansion states

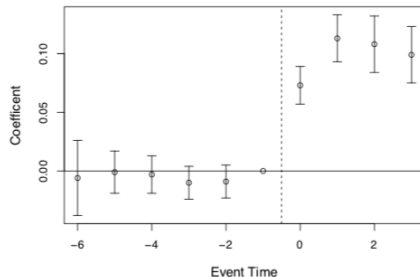


(a) Medicaid Eligibility

Did the policy actually move people onto Medicaid? Yes — enrollment rose sharply in expansion states post-2014, flat in non-expansion.

Without bite, there's no story. The mandate has to actually do something before we can attribute downstream changes to it.

Bite #2: any-insurance coverage rose

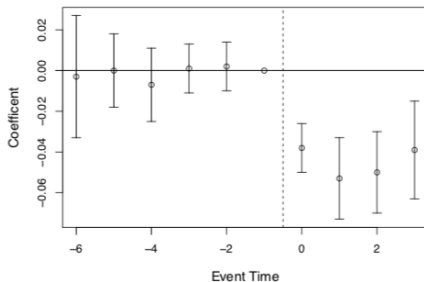


(b) Medicaid Coverage

Did the new Medicaid enrollees actually gain coverage they didn't have? Or did they just shift from private to Medicaid?

Any-insurance coverage rose — the expansion brought in the previously uninsured, not the already-insured.

Bite #3: the uninsured rate fell



(c) Uninsured

The mirror image of Bite #2.** Uninsured rate dropped sharply in expansion states post-2014. Three measures (Medicaid up, any-insurance up, uninsured down) converge on the same conclusion: **the policy bit.

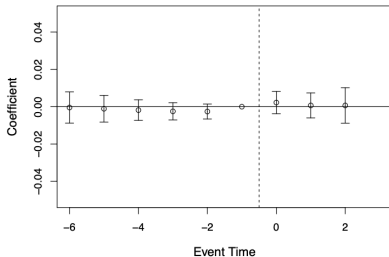
Five pieces of a strong DiD — now falsify

1. **Bite** — the policy moved what it was supposed to move. (*established.*)
2. **Event study** — pre/post coefficients around treatment.
3. **Falsification** — run a rival theory on a placebo group or outcome and try to reject *it*.
4. **Main result** — the DiD on the actual outcome.
5. **Mechanism** — the causal pathway from D to Y .

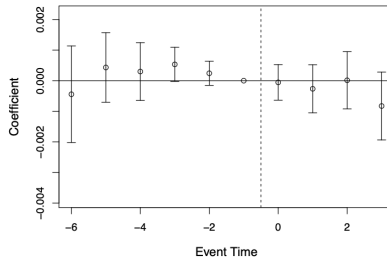
The rival: “some 2014 shock hit expansion states.” The falsification: run the same model on a group the policy shouldn’t reach.

Falsification: same model, on the 65+ population — no effect

Age 65+ in 2014



(c) Medicaid Coverage



(d) Annual Mortality

Same states, same years, same model — but run on the 65+ population. *They all have Medicare, so Medicaid expansion shouldn't touch their mortality.*

A rival theory ("some 2014 shock hit expansion states") would predict a mortality effect here. It didn't appear.

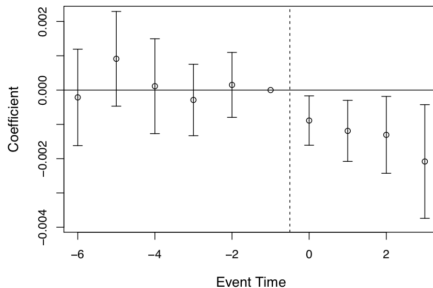
Rival rejected.

Five pieces of a strong DiD — the event study and main result

1. **Bite** — established.
2. **Event study** — pre/post coefficients around treatment.
3. **Falsification** — rival rejected.
4. **Main result** — the DiD on the actual outcome.
5. **Mechanism** — the causal pathway from D to Y .

Now we look at the outcome that matters: mortality. The event study is the main result, viewed in motion.

The main result: $\approx 9.2\%$ reduction in near-elderly mortality



- Post-2014: mortality declines in expansion states. $\approx 9.2\%$ reduction.
- Pre-period dots near zero, but **CIs are wide** — you cannot reject a straight line trending downward (*Rambachan & Roth 2024* sensitivity).
- The other four pieces are why the editors believed it. A critic now has to explain bite *and* falsification *and* the dynamic post-trend — not just one thing.

Five pieces of a strong DiD — and the fifth: mechanism

1. **Bite** — established.
2. **Event study** — established.
3. **Falsification** — rival rejected.
4. **Main result** — 9.2% reduction in near-elderly mortality.
5. **Mechanism** — the causal pathway from D to Y .
 - Mechanisms are the part of a result that's **transportable** — they extend across contexts where the same pathway operates.
 - Reduced-form top-line results often combine *several* mechanisms, which makes interpretation harder.
 - Here: disease-related deaths fall and compound, consistent with treated chronic conditions.

The five pieces, all five present in Medicaid

Bite	Three figures: Medicaid take-up up, any-insurance up, uninsured down
Event study	Pre-trends flat; post-trends decline in expansion states
Falsification	No effect on elderly (65+) — same states, same years, same model
Main result	$\approx 9.2\%$ reduction in mortality among the near-elderly
Mechanism	Disease-related deaths fall and compound; consistent with treated chronic conditions

*None of these alone is the case. **Together they are powerful.***

What we've done today — and tomorrow

Today:

1. DiD's empirical roots — Semmelweis, Ashenfelter.
2. DiD = calculation + assumption: the 2×2 and parallel trends.
3. Five regressions, one number.
4. Decomposing the 2×2 — the +0 trick and three failure cases.
5. Card and Krueger: parallel trends visualized.
6. Standard errors — vanilla, robust, cluster.
7. Population weighting: pick a target parameter.
8. Event studies as falsifications, by hand and via OLS.

Tomorrow: when the 2×2 stops working — covariates, staggered adoption (Goodman-Bacon, Callaway–Sant'Anna, Sun–Abraham, BJS, dCdH).